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Homework 6

1. What is a side-channel attack, and why are FPGAs susceptible to this type of attack?

A side-channel attack is an attack that does not directly break the algorithm itself, but rather it tries to recover secret information by noticing the physical leakage that is produced during operation. This leakage can include power consumption, electromagnetic emissions, timing behavior or other physical signals. FPGAs are susceptible to this type of attack because they implement cryptographic functions in their hardware and those operations can produce measurable leakage during execution. Since FPGA designs are programmable and widely used in security sensitive systems, attackers can study their power or EM behavior to try to recover the cryptographic keys or other sensitive information (Bhunia & Tehranipoor, 2019).

2. List two methods for mitigating hardware Trojans in FPGA designs.

Two methods for mitigating hardware Trojans in FPGA designs are bitstream encryption and bitstream obfuscation. Bitstream encryption helps protect the FPGA configuration from being able to be read or modified by an attacker, which makes Trojan insertion more difficult to achieve. Bitstream obfuscation helps by adding another layer of protection by making the mapping between the bitstream and the actual FPGA resources harder to interpret, this helps reduce the chance of reverse engineering or malicious modification. These methods are important because FPGA security depends on protecting the configuration bitstream from piracy, tampering and unauthorized changes (Bhunia & Tehranipoor, 2019).

3. Briefly explain how fault injection attacks can be used to compromise FPGA security.

Fault injection attacks can be used to compromise FPGA security by using power adjustment, radiation or optical fault injection. This is done to alter the configuration bits or disrupt the security related computations. Attackers can then bypass protection mechanisms, trigger malfunctions, corrupt the mapped design or weaken the reliability of the FPGA system. In SRAM-based FPGAs especially, changing even individual configuration bits can create serious security problems because the bit stream is what directly controls the hardware functionality (Bhunia & Tehranipoor, 2019).

4. What is the role of bitstream encryption in securing FPGA designs?

The role of bitstream encryption in FPGA designs is to protect the configuration bitstream from unauthorized access, copying and modifications. Since the bitstream defines how the FPGA hardware behaves, encrypting it helps prevent attackers from being able to understand the design, reverse engineer it or tamper with it to insert malicious logic. Bitstream encryption is therefore critical and provides protection against threats such as IP piracy and some forms of malicious modification. However, its security still depends on protecting the encryption key because if the key is ever exposed then the encrypted bitstream is compromised (Bhunia & Tehranipoor, 2019).

References

Bhunia, S., & Tehranipoor, M. H. (2019). *Hardware security: A hands-on learning approach*. Morgan Kaufmann.